

Edge Brain-Tumour Detection

CNN + GRU + PoSTAL | Streamlit app | README and requirements

Project: Lightweight CNN Architecture for Real-Time Brain Tumour Detection on Edge Devices

1. Quick start

```
pip install -r requirements.txt
streamlit run app.py
```

The app works immediately with a **Demo engine** (NumPy only, no TensorFlow needed). The synthetic dataset is in data/ and is regenerated automatically if missing.

2. Train the real model (needs TensorFlow 2.16 or newer)

```
python synthetic_data.py --n 500 --out data # optional: regenerate dataset
python train.py --epochs 30                # saves weights, metrics, TFLite to models/
streamlit run app.py                       # sidebar offers 'Trained Keras model'
```

Note: the Keras model (model.py, train.py) has not yet been run, so no accuracy figure exists for it until you train it. The Demo engine is a hand-built stand-in tuned on the synthetic data; its scores are not evidence of real-world accuracy.

3. Files

File	Purpose
app.py	Streamlit UI: Overview, Dataset Explorer, Architecture, Detect, Evaluation, Edge Benchmark
synthetic_data.py	Synthetic 3-D MRI volume generator (tumours, distractors, noise)
pipeline_np.py	Demo engine: non-learned stand-in mirroring CNN, GRU and PoSTAL stages
model.py	Keras model: depthwise-separable CNN, Bi-GRU over slices, PoSTAL layer
train.py	Training, test evaluation, TFLite export and latency
data/synthetic_mri.npz	Volumes (N,8,64,64) uint8, masks, labels
data/metadata.csv	Per-volume info and train/val/test split
requirements.txt	Python dependencies

4. Moving to real data (BraTS / Figshare / Br35H)

Prepare arrays in the same layout (volumes (N,S,H,W) uint8, masks, labels) and a metadata.csv with columns volume_id, label, tumour_type, split (plus any others), then point train.py --data at that folder.

5. requirements.txt

```
streamlit>=1.35
numpy
pandas
scipy
scikit-learn
matplotlib
pillow
# Only needed for training / the 'Trained Keras model' engine
# (the app runs without it in demo mode):
tensorflow>=2.16
```

Research prototype only - not a medical device and not for clinical use.